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Molecular Spin Qubit Relaxation: Beyond Perturbation and the Spin-Phonon Polaron

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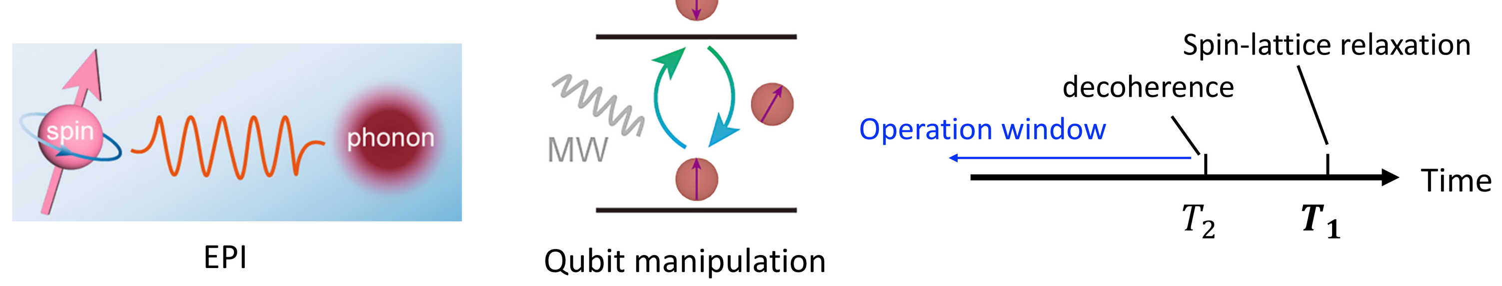
Introduction

- Understanding **relaxation and decoherence** in spin quantum devices requires a rigorous treatment of **spin-phonon interaction** (SPI):



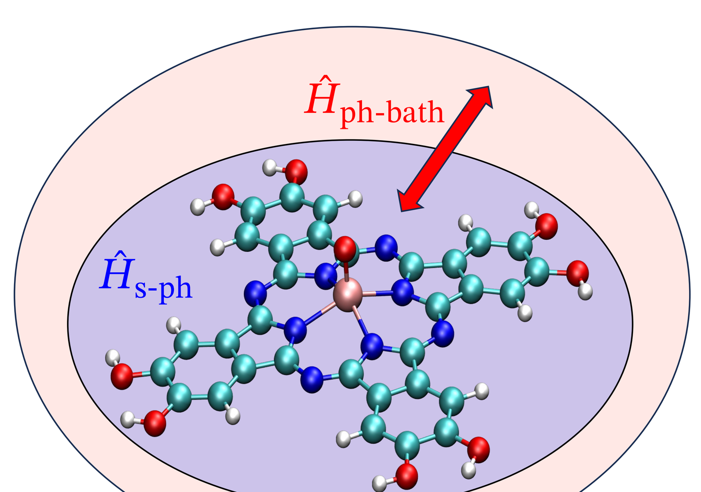
Freedman, D. E. *Chem. Sci.* **2022** 13 (23); Sun, L *Inorg. Chem. Front.*, **2024**,11, 8660-8670

- **Quadratic SPI**, involving two-phonon scattering, is the key driver of spin-lattice relaxation in many materials. While perturbative models (e.g., Raman relaxation) are widely used, they fail in regimes of strong coupling.
- We developed both **perturbative and exact theoretical framework** for quadratic SPI. This allows us to move beyond perturbation theory and ask: *What are the signatures of strong quadratic coupling?*

Theory

- Model Hamiltonian

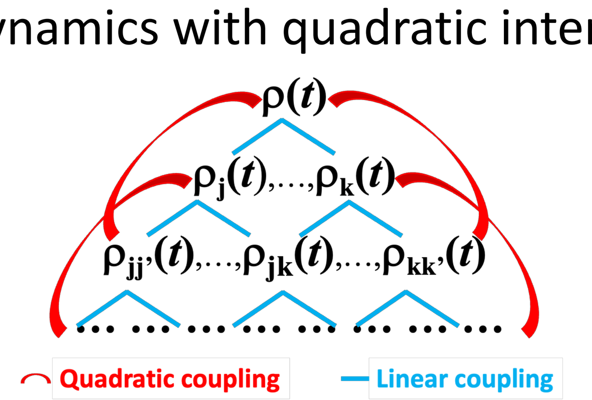
$$\hat{H} = \hat{H}_{\mathrm{s}} + \hat{H}_{\mathrm{ph}} + \hat{H}_{\mathrm{s\text{-}ph}} + \hat{H}_{\mathrm{ph\text{-}bath}},$$
$$\hat{H}_{\mathrm{s}} = \mu_{\mathrm{B}}\,\mathbf{B}\,\mathbf{g}\,\hat{\mathbf{S}},$$
$$\hat{H}_{\mathrm{ph}} = \sum_i \frac{\Omega_i}{2}(\hat{P}_i^2 + \hat{Q}_i^2),$$
$$\hat{H}_{\mathrm{ph\text{-}bath}} = \sum_i\sum_j \frac{\omega_{ij}}{2}\Big[\hat{p}_{ij}^2 + \Big(\hat{q}_{ij} - \frac{c_{ij}}{\omega_{ij}}\hat{Q}_i\Big)^2\Big],$$
$$\hat{H}_{\mathrm{s\text{-}ph}} = \Big(\sum_i g_i^{(1)}\hat{Q}_i + \sum_{i,j} g_{ij}^{(2)}\hat{Q}_i\hat{Q}_j\Big)\hat{\mathbf{S}},$$
$$\mathbf{g}_i^{(1)} \equiv \mu_{\mathrm{B}}\mathbf{B}\frac{\partial \mathbf{g}}{\partial Q_i}\Big|_{Q=0}, \quad \mathbf{g}_{ij}^{(2)} \equiv \mu_{\mathrm{B}}\mathbf{B}\frac{\partial^2 \mathbf{g}}{\partial Q_i \partial Q_j}\Big|_{Q=0},$$



Garg-Onuchic-Ambegaokar (GOA) mapping

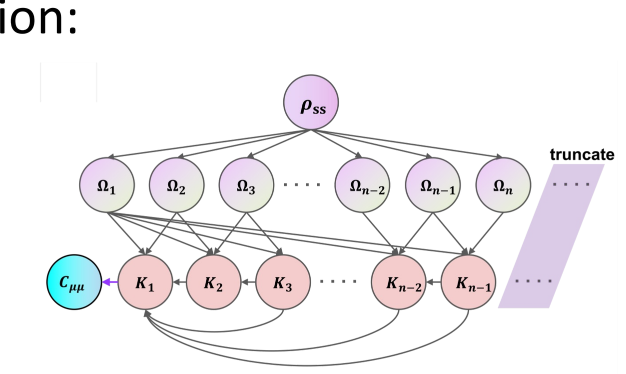
$$\hat{H}^{\mathrm{eff}} = \hat{H}_{\mathrm{s}} + \sum_i\sum_k \frac{\tilde{\omega}_{ik}}{2}(\tilde{p}_{ik}^2 + \tilde{q}_{ik}^2) + \Big[\sum_i \mathbf{g}_i^{(1)}\Big(\sum_k \tilde{c}_{ik}\tilde{q}_{ik}\Big) + \sum_{i,j}\mathbf{g}_{ij}^{(2)}\Big(\sum_k \tilde{c}_{ik}\tilde{q}_{ik}\Big)\Big(\sum_{k'} \tilde{c}_{jk'}\tilde{q}_{jk'}\Big)\Big]\hat{\mathbf{S}}.$$

- Dynamics with quadratic interaction:



Extended DEOM

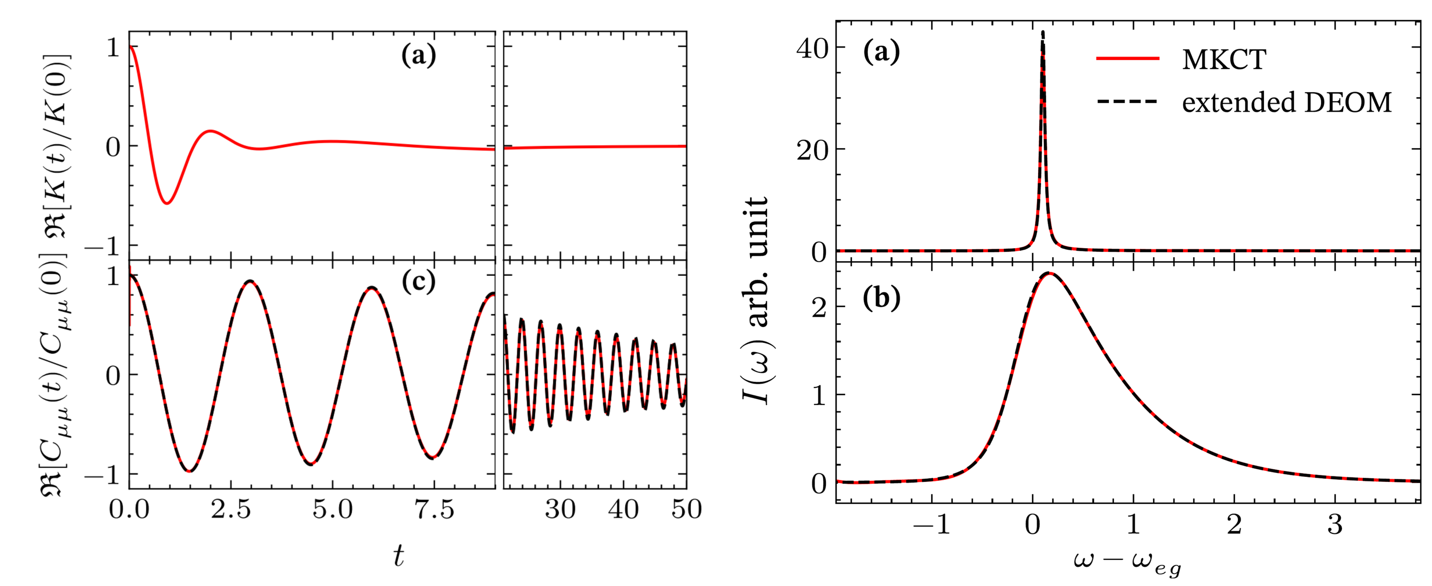
Rui-Xue Xu, ..., YiJing Yan. *JCP* **2018** 148, 114103



MKCT

W. Liu, ..., W. Dou, *PRL* **2025** 135 (14)

e-DEOM	MKCT
A statistical quasi-particle theory for OQS. Extended to quadratic. Not validated.	**An effecient theory for memory kernel of GQMEs in OQS. Exact if the input moments are exact.**



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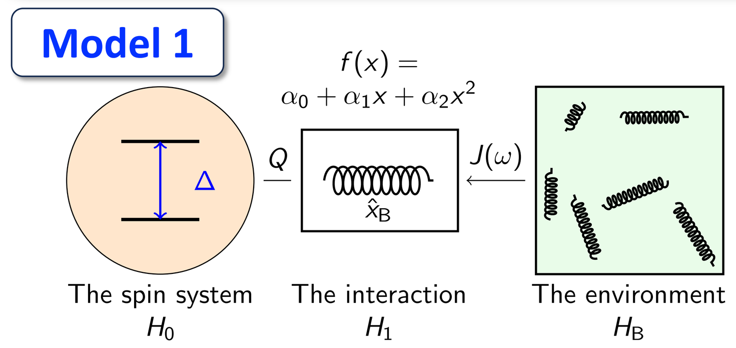
- Relaxation rate from kinetic rate kernel:

$$\mathscr{P} \equiv \sum_i \mathscr{P}_i, \quad \mathscr{P}_i \equiv |i\rangle\langle i| \qquad k_{f\leftarrow i} = \int_0^\infty \mathrm{d}\tau\, \mathcal{K}_{f\leftarrow i}(\tau)$$
$$\mathcal{K}_{f\leftarrow i}(t) = \langle\langle f|\mathscr{P}\mathrm{i}\mathcal{L}e^{-(\mathbf{I}-\mathscr{P})\mathrm{i}\mathcal{L}}(\mathbf{I}-\mathscr{P})\mathrm{i}\mathcal{L}|i\rangle\rangle \qquad T_1^{-1} = k_{1\leftarrow 0} + k_{0\leftarrow 1}$$

Strong quadratic SPI effect in spin-lattice relaxation

Model 1



Spin-1/2 + 1 vibration mode

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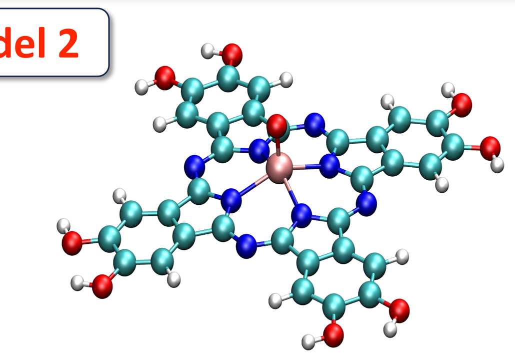
Model 2



Spin-1/2 + 168 vibration modes (VOPc)

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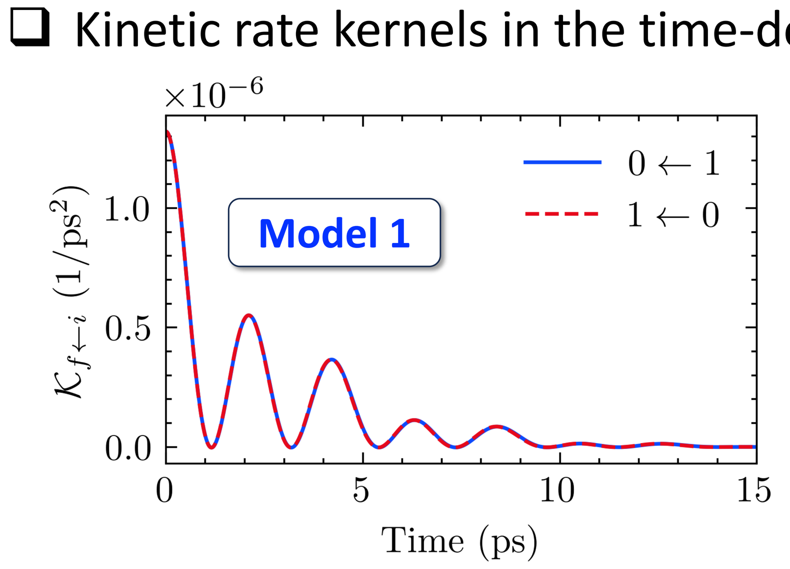
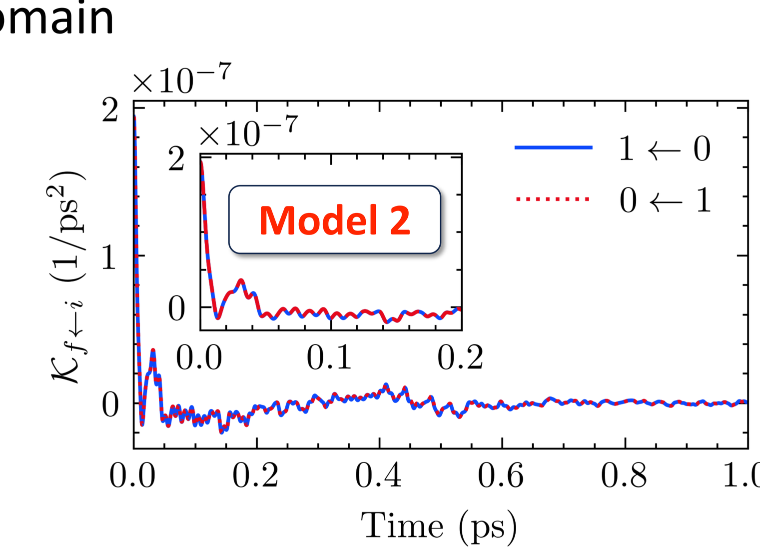
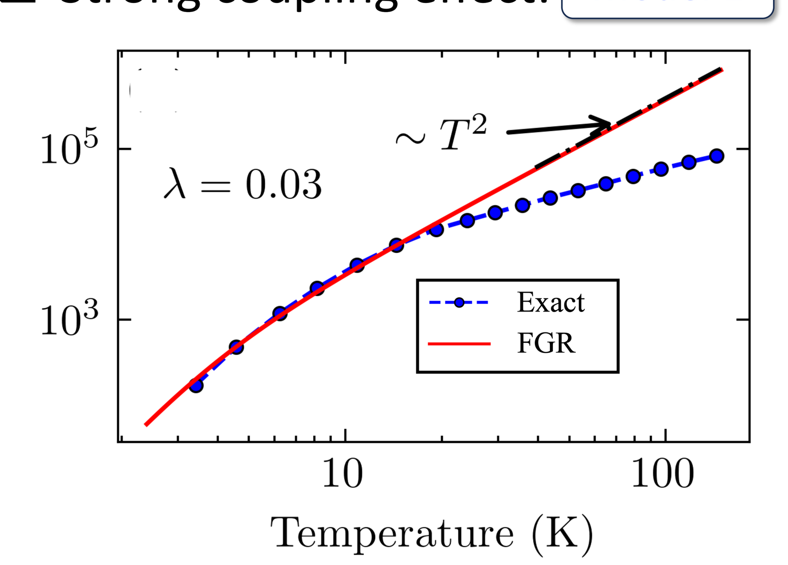
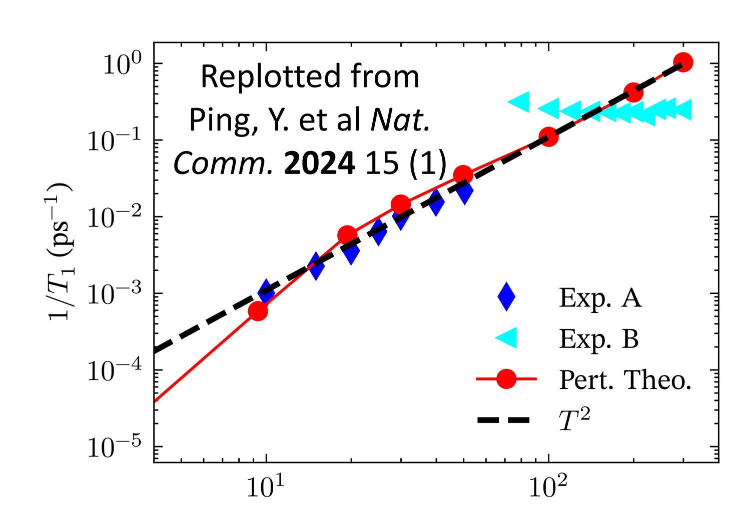
- Kinetic rate kernels in the time-domain



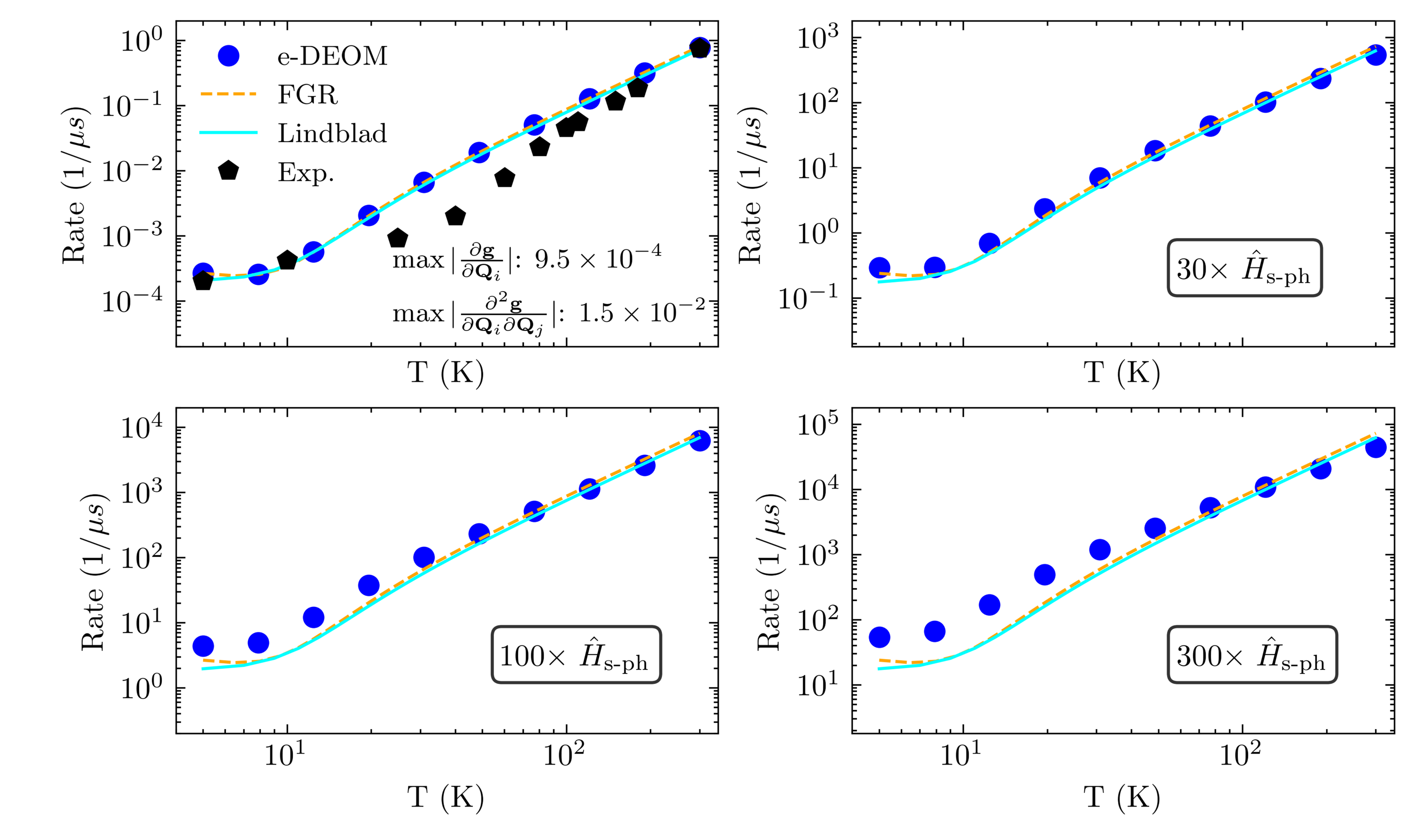


- Strong coupling effect: **Model 1**



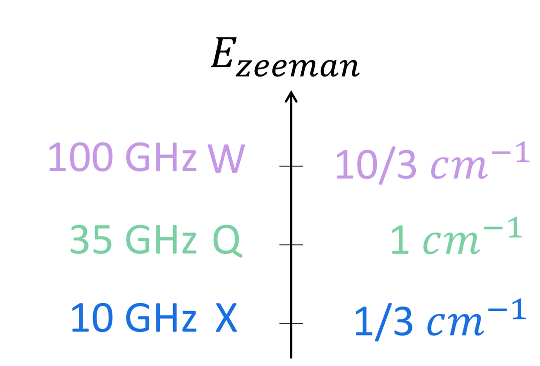


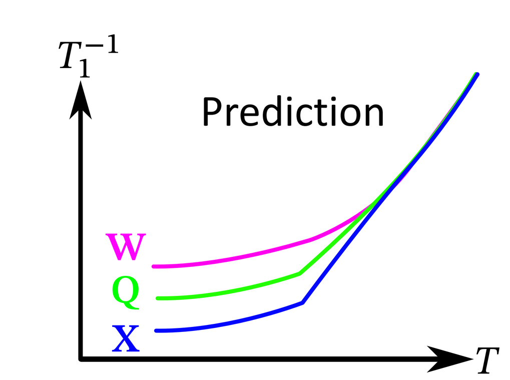
- (Artificial) strong coupling effect: **Model 2**

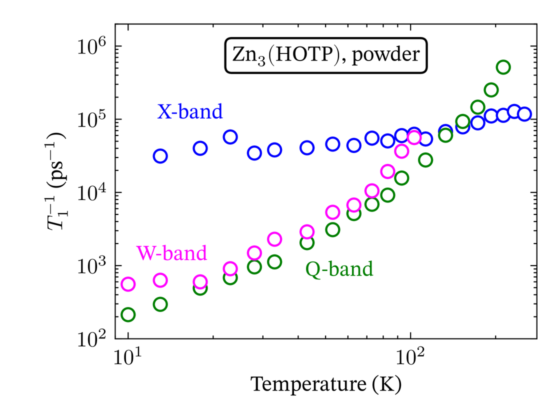


Tunable spin-lattice relaxation in ZnHOTP

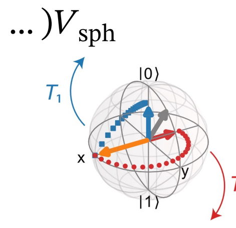
- Unconventional Spin relaxation in ZnHOTP (chiral)





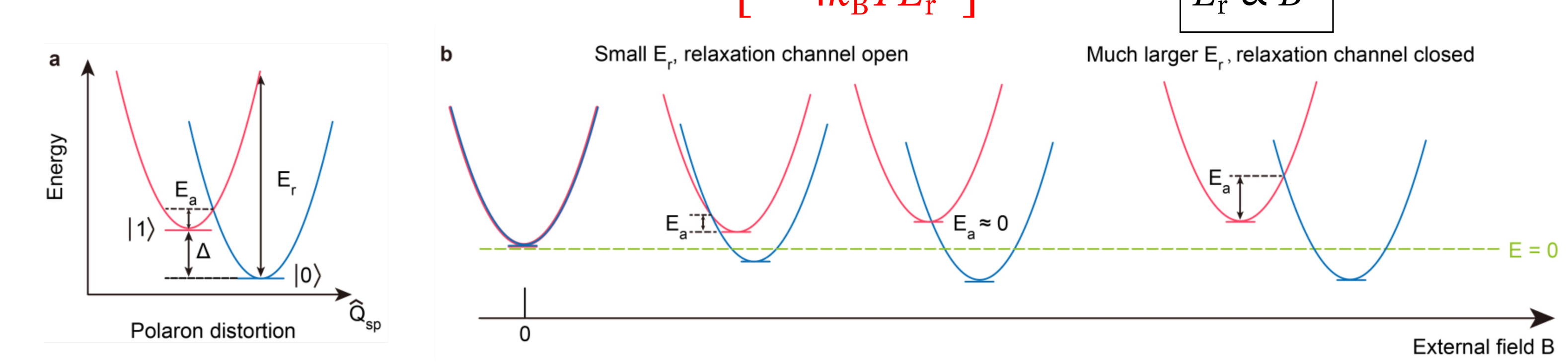


- We introduce a spin-phonon polaron model:

Traditional picture	Spin-phonon polaron picture
$H_{\mathrm{sph}} = (\alpha_1\hat{Q} + \alpha_2\hat{Q}^2 + \ldots)V_{\mathrm{sph}}$ $V_{\mathrm{sph}} \propto \begin{cases}\sigma_x & \text{Relaxation, } T_1.\\ \sigma_z & \text{Decoherence, } T_2\end{cases}$  Only consider transverse coupling: $V_{\mathrm{sph}} \propto \sigma_x$	$H_{\mathrm{sph}} = (\alpha_1\hat{Q}_{\mathrm{r}} + \alpha_2\hat{Q}_{\mathrm{r}}^2)\sigma_x + \gamma_{\mathrm{p}}\hat{Q}_{\mathrm{sp}}\sigma_z$ "relaxation" modes "modulation modes" Small polaron transform $\tilde{H}_{\mathrm{sph}} = (\alpha_1\hat{Q}_{\mathrm{r}} + \alpha_2\hat{Q}_{\mathrm{r}}^2)\exp\Big(\mathrm{i}\sum_k \tilde{\lambda}_k\hat{p}_{k,\mathrm{sp}}\Big)\sigma_+ + \mathrm{h.c.}$

Result: a Marcus-theory-like result (valid if strong modulation)

$$T_1^{-1}(\text{with polaron}) \propto T_1^{-1}\exp\Big[-\frac{(\Delta - E_{\mathrm{r}})^2}{4k_{\mathrm{B}}TE_{\mathrm{r}}}\Big] \qquad \boxed{\begin{matrix}\Delta \propto B,\\ E_{\mathrm{r}} \propto B^2\end{matrix}}$$



A. Zhou, R.-H. Bi, ..., W. Dou, L. Sun, arXiv:2506.04885 **2026**

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